

PURE: Refining Feed-forward 3D Reconstruction with Unlabeled Post-training

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Abstract

Feed-forward 3D reconstruction models have achieved remarkable progress through large-scale training and strong learned priors. However, they remain fragile when applied to dynamic in-the-wild videos, where non-rigid motion, occlusions, and illumination changes are prevalent. A key limitation lies in their reliance on expensive depth and camera pose annotations, which hinders scaling to diverse real-world dynamic data and restricts generalization. To address this issue, we propose **PURE**, an unlabeled post-training refinement framework that improves depth and camera estimation without requiring ground-truth supervision. Instead of relying

on annotations, PURE elegantly distills robust structural knowledge from state-of-the-art 2D priors into 4D feed-forward architectures. PURE leverages relative depth constraints, affine-invariant geometry distillation, and pose refinement based on feature correspondences to stabilize geometric predictions under challenging dynamic conditions. Extensive experiments on both synthetic and real-world dynamic benchmarks demonstrate that PURE delivers consistent improvements across multiple state-of-the-art architectures. Furthermore, our data-efficient paradigm effectively mitigates domain shifts, successfully unlocking the architectural limits of these base models in challenging, in-the-wild dynamic scenes.



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CCS Concepts

• Computing methodologies → 3D imaging.

Keywords

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1 Introduction

Recovering accurate 3D geometry and camera motion from videos is a fundamental problem in computer vision, with applications [2, 11, 13, 49, 50] ranging from robotics and autonomous driving to augmented reality and digital content creation. Classical Structure-from-Motion (SfM) and visual SLAM systems achieve high accuracy through iterative optimization over sparse correspondences and bundle adjustment, but rely on rigid-scene assumptions, sufficient parallax, and carefully engineered pipelines. These limitations compromise the robustness of such methods in dynamic scenes and impede their scalability to unconstrained real-world videos.

Recent advances in feed-forward 3D reconstruction models have significantly shifted this paradigm. By leveraging large-scale datasets such as CO3Dv2 [28] and powerful Transformer-based [36] neural architectures, these models directly regress depth and camera parameters from images or videos, enabling fast and scalable reconstruction. Large video-based models such as VGGT [37], Pi3 [42], and DepthAnything3 [21] demonstrate impressive zero-shot performance on long sequences, highlighting the potential of data-driven priors for 3D understanding. Despite this progress, most feed-forward models are trained with strong supervision, including ground-truth depth, camera poses, or synthetic renderings, which are costly to acquire and limit generalization to real-world dynamic scenes.

Adapting pre-trained 3D reconstruction models to in-the-wild videos without annotations remains a challenging problem. Existing unsupervised approaches (e.g., [10, 53]) typically rely on photometric consistency, which fundamentally degrades in the presence of illumination changes, non-Lambertian surfaces, and highly dynamic objects. Conversely, methods relying on test-time optimization or per-scene adaptation [12, 20, 25, 47] incur prohibitive computational overhead, rendering them impractical for efficient video processing. Consequently, an effective annotation-free strategy that refines feed-forward models at the dataset level—preserving inference efficiency—remains largely underexplored.

In this work, we propose a novel post-training framework, termed **PURE** (Post-training Unlabeled REconstruction), for annotation-free refinement of feed-forward 3D reconstruction models. PURE adapts a pre-trained base model using unconstrained video sequences, without requiring ground-truth supervision or slow test-time optimization. Instead of relying on photometric signals, PURE distills robust structural knowledge from state-of-the-art 2D monocular priors into the 4D feed-forward architecture. To enable stable refinement despite the absolute scale and shift errors inherent in teacher models, we introduce relative depth constraints that preserve ordinal depth, local affine-invariant geometry distillation that maintains fine-grained surface structure, and correspondence-based pose refinement that enforces multi-view consistency.

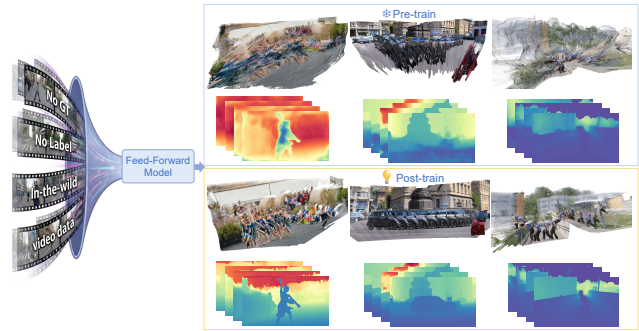


Figure 1: Our annotation-free post-training significantly improves feed-forward 3D reconstruction in dynamic scenes, yielding more accurate and temporally consistent geometry compared to the original pre-trained models.

We validate our approach through extensive experiments on both synthetic and real-world dynamic datasets, including video depth estimation, monocular depth prediction, and camera pose evaluation. Across multiple state-of-the-art architectures, our post-training strategy consistently improves performance and robustness, particularly in challenging dynamic scenes. These results demonstrate that annotation-free post-training offers a practical and scalable pathway for advancing feed-forward 3D reconstruction in real-world settings.

In summary, the contributions of this work are as follows:

- **A Novel Post-Training Paradigm:** We propose PURE, a unified, model-agnostic post-training framework that adapts pre-trained feed-forward 3D reconstruction models to in-the-wild videos using purely unlabeled data, eliminating the need for ground-truth annotations or test-time optimization.
- **Robust Geometry Distillation:** We introduce a noise-tolerant distillation strategy utilizing relative depth ordering and local affine-invariant constraints. This allows the base model to absorb rich structural priors from 2D teacher models while actively filtering out their absolute scale and shift errors.
- **Dynamic-Aware Pose Refinement:** We design a correspondence-based camera pose refinement mechanism featuring a rigorous filtering and sampling pipeline, ensuring reliable multi-view consistency even in scenes dominated by non-rigid dynamic motion.

2 Related Work

2.1 Optimization-based 3D Reconstruction

Classical SfM and visual SLAM systems recover camera poses and scene geometry through iterative optimization using 2D correspondences and bundle adjustment. While accurate in controlled settings (e.g., COLMAP [31, 32]), they rely on rigid scenes, sufficient parallax, and reliable feature matches.

Recent optimization-based methods [6, 12, 20, 25, 35, 43, 44, 51] extend this paradigm to handle dense geometry and dynamic content. Systems such as MegaSAM [20] and ViPE [12] combine dense depth estimation, motion reasoning, and iterative refinement to reconstruct long video sequences. Despite improved robustness, these

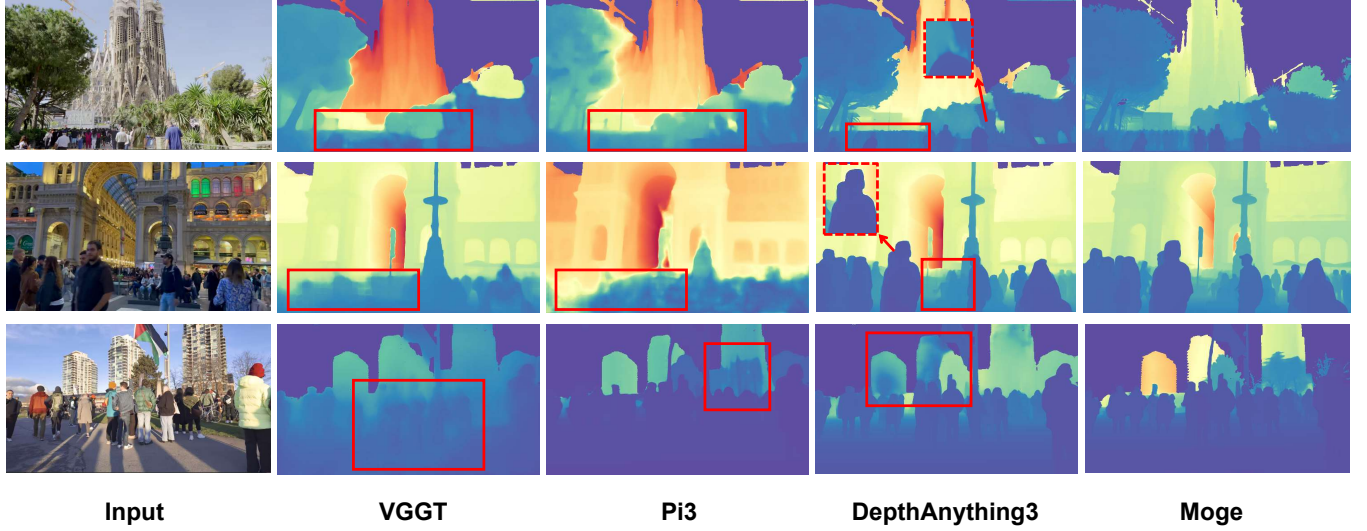


Figure 2: MoGe provides accurate frame-wise relative geometry, while feed-forward 3D models ensure temporal scale consistency. We unify both by transferring monocular geometric cues while preserving temporal consistency.

approaches remain computationally expensive and require careful system design, limiting their scalability to diverse in-the-wild videos.

2.2 Feed-forward 3D Reconstruction

To overcome the limitations of iterative optimization, recent works [5, 7, 14, 17, 18, 21, 24, 37, 39, 41, 42, 45, 46, 48, 55] have explored feed-forward 3D reconstruction models leveraging large-scale data-driven priors. These methods can be broadly divided into pair-wise and video-based approaches.

Pair-wise methods originate from DUST3R [41], which formulates multi-view reconstruction as dense point-map regression from image pairs. Subsequent works [14, 18, 24, 48] extend this paradigm to dynamic videos by incorporating temporal fine-tuning or monocular depth priors, but remain limited by pair-wise processing and poor scalability to long sequences.

Video-based methods extend feed-forward reconstruction to full video sequences. Models such as CUT3R [39], StreamVGGT [55], Point3R [45], and TTT3R [7] capture long-range temporal dependencies via memory or streaming mechanisms, while VGGT [37], Pi3 [42], and DepthAnything3 [21] learn strong spatio-temporal priors from large-scale video data. Despite strong performance, these models remain brittle on dynamic, in-the-wild videos without annotations, motivating annotation-free post-training refinement such as ours.

3 Methodology

We propose a unified framework termed **PURE** for unlabeled post-training of feed-forward 3D reconstruction models (Fig. 3). Given a sequence of K images $I = \{I_1, \dots, I_K\}$, the base model $\mathcal{F}_\theta$ predicts camera poses $\mathbf{T}$, intrinsics $\mathbf{K}$, and depth or point maps ($\mathbf{D}/\mathbf{P}$). These predictions are refined via local geometry optimization using ordinal depth, affine-invariant constraints, and intrinsic distillation, and through dense correspondences from GIM [33] filtered

by confidence, cycle consistency, and inverse density sampling. By combining these mechanisms, our framework improves both geometry and camera pose estimation, yielding more accurate and temporally consistent 3D reconstructions from unlabeled video sequences.

3.1 Relative Local Geometry Optimization

Monocular depth estimators [40] produce depth maps with more accurate relative ordering and local geometric structures compared to feed-forward 3D reconstruction models [21, 37, 42], as illustrated in Fig. 2. However, their predictions often exhibit temporal scale inconsistencies. Directly supervising feed-forward 3D reconstruction models with monocular depth can lead to pronounced “layering” artifacts, since monocular object-wise depth scales may vary significantly across frames and deviate from the 3D reconstruction model’s learned geometric prior. To address this issue, we avoid absolute depth supervision and instead introduce two relative constraints: an ordinal depth ranking loss and a local affine-invariant loss. These losses transfer the reliable structural and local geometric cues from the monocular teacher while remaining invariant to global scale and shift, enabling stable refinement without collapsing the underlying geometry of the 3D reconstruction models.

Ordinal Depth Constraint. For an input image I_t , let D_t be the predicted depth map and $\hat{D}_t = \mathcal{T}_{mono}(I_t)$ be the pseudo-ground truth generated by a monocular teacher. Inspired by MoDGS [22], we enforce consistency in the relative order of depth values between pixel pairs, making the optimization robust to absolute scale fluctuations. We define the robust ordinal loss as:

$$\mathcal{L}_{order} = \sum_{(u_1, u_2) \in \mathcal{P}} \left\| \tanh(\alpha(D_t(u_1) - D_t(u_2))) - R(\hat{D}_t(u_1), \hat{D}_t(u_2)) \right\|_1 \quad (1)$$

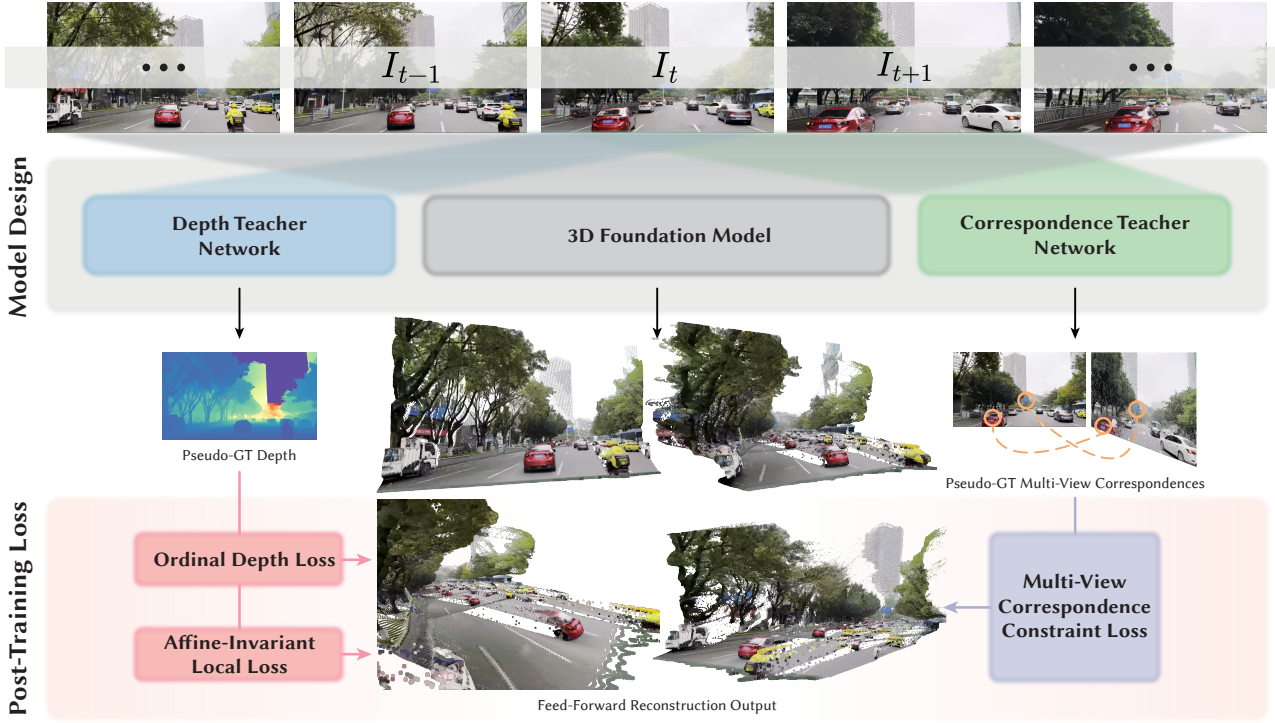


Figure 3: Overview of PURE, our annotation-free post-training framework. Given an unlabeled image sequence, a pre-trained feed-forward model predicts depth and camera poses. We refine the predictions via two complementary branches: (1) local geometry refinement using relative depth and affine-invariant constraints with intrinsic distillation; and (2) pose refinement using robust correspondence-based constraints from GIM matches, enforcing cross-frame geometric consistency.

where $\mathcal{P}$ represents a set of sampled pixel pairs, α is a scaling factor, and $R(\cdot, \cdot)$ is an indicator function returning $\{+1, -1\}$ based on the depth ranking of the teacher model. This constraint preserves structural integrity without propagating erroneous absolute scales.

Intrinsic Parameter Distillation. Estimating camera intrinsics—particularly the focal length—from unconstrained videos is inherently ill-posed due to the strong ambiguity between depth scale and intrinsics. In practice, we observe that directly aligning 3D point clouds predicted by monocular depth estimators (e.g., MoGe [40]) and feed-forward 3D reconstruction models is problematic: discrepancies in the estimated intrinsics result in mismatched ray directions, i.e., the same image pixels are back-projected along different viewing rays, introducing structural distortions that make the resulting point clouds incompatible under simple scale-and-shift 3D alignment.

Interestingly, we find that pre-trained 3D reconstruction models such as VGGT [37] and Pi3 [42] tend to produce relatively stable intrinsic parameters or ray directions, even when their depth estimates are inaccurate. Motivated by this observation, we design a distillation strategy that explicitly decouples ray geometry from depth magnitude.

Building on this insight, we employ a distillation-based strategy to generate pseudo 3D local points. Let $\mathbf{p}$ denote the homogeneous pixel coordinates. We combine the robust ray directions $\mathbf{r}_{pretrain}$,

derived from the pre-trained model’s intrinsics $\mathbf{K}_{pre}$, with the corresponding depth values d_{mono} from the monocular depth estimator. This process is formulated as:

$$\mathbf{r}_{pretrain} = \mathbf{K}_{pre}^{-1} \mathbf{p} \quad (2)$$

$$\mathbf{P}_{pseudo} = d_{mono} \cdot \mathbf{r}_{pretrain} \quad (3)$$

These synthesized 3D points, $\mathbf{P}_{pseudo}$, serve as supervisory signals that implicitly regularize the model’s estimation of camera intrinsics, without requiring explicit labels.

Local Geometry Constraint. To capture high-frequency surface details, we adopt the Affine-Invariant Distillation Loss from MoGe [40]. Instead of aligning the entire image globally—which is sensitive to outliers—we enforce geometric consistency within local spherical regions. For a local region centered at pixel p_j , we align the predicted point map $\mathbf{P}$ to the teacher’s pseudo point map $\mathbf{P}_{pseudo}$, computed as in Eq. (3), by solving for the optimal scale s and shift $\mathbf{t}$:

$$\mathcal{L}_{local} = \sum_j \min_{s, \mathbf{t}} \sum_{i \in \mathcal{N}(p_j)} \|\mathbf{P}_{pseudo_i} - (s \cdot \mathbf{P}_i + \mathbf{t})\|_1 \quad (4)$$

This formulation preserves local curvature and fine-grained surface details even under global scale drift.

3.2 Pose Estimation via Correspondences

Accurate camera pose estimation is essential for coherent multi-view reconstruction. Traditional Structure-from-Motion (SfM) pipelines rely on sparse 2D feature correspondences to recover epipolar geometry. To approximate this process in an unsupervised manner, we leverage GIM [33] for robust correspondence estimation between image pairs. These correspondences serve as powerful constraints, ensuring that matched 2D pixels correspond to the same 3D point in space. To guarantee the reliability of these constraints, we implement a rigorous filtering pipeline:

- **Dynamic and Confidence Filtering:** Matches located in dynamic regions or with confidence scores below a pre-defined threshold τ are discarded.
- **Cycle Consistency Check:** We enforce forward-backward cycle consistency ($p_A \rightarrow p_B \rightarrow p'_A$) to remove ambiguous or erroneous matches.
- **Inverse Density Sampling (IDS):** High-confidence correspondences often cluster in texture-rich regions, potentially biasing the global pose estimation. To address this, we employ IDS, which prioritizes sampling matches from sparser, lower-confidence regions. This ensures a spatially uniform distribution of constraints and improves global robustness.

3.3 Model Training

We fine-tune three representative feed-forward 3D reconstruction models: VGGT [37], Pi3 [42], and DepthAnything3 [21]. For VGGT and DepthAnything3, only a subset of intermediate layers is updated to balance adaptation capability with training stability. For Pi3, we observe severe convergence instability and grid-like artifacts during fine-tuning. We attribute this to its pixel-shuffle-based point head, whose inherent grid structure is amplified by our local geometry constraint. To mitigate this, we replace the original point head with a multi-level convolutional feature head, producing smoother and more stable point predictions. The new head is jointly trained during post-training.

A key challenge in dynamic scene reconstruction is the lack of high-quality real-world training data. Most existing dynamic datasets are synthetic, while real-world datasets such as Waymo [34] provide sparse LiDAR depth, which is insufficient for learning fine-grained geometry. To address this limitation, we leverage large-scale unlabeled videos from real dynamic scenes. As summarized in Tab. 1, we utilize high-quality sequences from SpatialVID-HQ [38] and Sekai-Real-Walking-HQ [19]. Pseudo supervision is obtained using off-the-shelf foundation models: MoGe [40] for monocular depth, Grounded SAM 2 [15, 16, 23, 27, 29, 30] for dynamic object and sky segmentation, and GIM Roma [33] for dense correspondence estimation.

To further prevent catastrophic forgetting, we augment training with synthetic datasets including OmniWorld [54], PointOdyssey [52], VKITTI [4], and BEDLAM [1], as well as real-world driving videos from Waymo. This combination of real and synthetic data enables robust post-training across diverse dynamic environments.

4 Experiment

We present implementation details in Sec. 4.1 and primarily evaluate our approach on dynamic-scene benchmarks, focusing on video

Table 1: Statistics of datasets for fine-tuning.

Name	#Scene	Avg. #Img	Total #Img	Type
SpatialVID	1.8k	~490	882k	Real(Unlabeled)
Sekai-Real	4k	900	3.6M	Real(Unlabeled)
OmniWorld	121	~4k	484k	Synthetic
VKITTI	100	425	42k	Synthetic
BEDLAM	30	250	7.5k	Synthetic
PointOdyssey	131	1.8k	240k	Synthetic
Waymo	1k	200	200k	Real

depth estimation (Sec. 4.2) and camera pose estimation (Sec. 4.3). To further validate the effectiveness of our annotation-free post-training strategy, we conduct additional experiments in which post-training is performed directly on the test sets (Sec. 4.4). This setting isolates the impact of the proposed post-training objectives from data scale and distribution effects, demonstrating that our method consistently improves geometric predictions even when adaptation is carried out on the evaluation data itself. Finally, we present detailed ablation studies in Sec. 4.5 to analyze the contributions of individual framework components, alongside a scaling-up analysis that investigates the impact of post-training data volume on overall performance.

4.1 Implementation Details

We fine-tune three baseline models: VGGT [37], Pi3 [42], and DepthAnything3 (DA3) [21]. For VGGT and DA3, optimization is applied to the intermediate global attention layers. For Pi3, in addition to these layers, we also train a newly replaced point prediction head.

To support video inputs with varying resolutions, we constrain image aspect ratios to the range [0.5, 2.0] and limit the total number of pixels to between 160k and 250k. Our training setup includes both multi-view and single-view inputs, where the number of views is uniformly sampled from [1, 24]. To ensure stable training efficiency, the batch size is dynamically adjusted to maintain an approximately constant number of tokens per iteration.

All experiments are conducted on 4× H800 GPUs, using dynamic batch sizing and a gradient accumulation step of 2. We adopt AdamW optimizer with a learning rate of 1×10^{-5} , weight decay of 5×10^{-2} , and $\beta = (0.9, 0.95)$. The learning rate is scheduled using OneCycleLR with cosine annealing and a maximum learning rate of 1×10^{-5} . Gradient clipping is applied with a maximum norm of 1.0.

In terms of training efficiency, the PURE post-training stage takes approximately 20 hours for VGGT (12K iterations) and 8 hours for DA3 (5K iterations). For Pi3, training is divided into two stages: a depth head initialization phase (10K iterations, approximately 12 hours), followed by a post-training phase (5K iterations, approximately 8 hours).

Notably, PURE is a post-training framework that introduces negligible architectural modifications (except for a minor head replacement in Pi3). As a result, the inference computational cost remains identical to that of the original base models.

Table 2: Video depth estimation on Sintel, Bonn, KITTI and Dycheck. Best results are highlighted.

Method	Params	Align	Sintel		Bonn		KITTI		DyCheck	
			Abs Rel ↓	$\delta < 1.25$ ↑	Abs Rel ↓	$\delta < 1.25$ ↑	Abs Rel ↓	$\delta < 1.25$ ↑	Abs Rel ↓	$\delta < 1.25$ ↑
VGGT	1.16B		0.279	0.657	0.058	0.969	0.062	0.971	0.134	0.827
Ours (VGGT)	1.16B		0.220	0.724	0.045	0.975	0.040	0.984	0.062	0.956
Pi3	959M	scale	0.230	0.674	0.049	0.975	0.038	0.986	0.048	0.980
Ours (Pi3)	989M		0.212	0.733	0.043	0.977	0.039	0.987	0.049	0.981
DA3	1.21B		0.236	0.702	0.047	0.974	0.058	0.975	0.053	0.959
Ours (DA3)	1.21B		0.224	0.710	0.045	0.975	0.048	0.978	0.040	0.982
VGGT	1.16B		0.215	0.704	0.050	0.971	0.053	0.969	0.113	0.852
Ours (VGGT)	1.16B	scale & shift	0.178	0.741	0.043	0.976	0.038	0.984	0.055	0.976
Pi3	959M		0.210	0.726	0.044	0.975	0.037	0.986	0.042	0.979
Ours (Pi3)	989M		0.195	0.750	0.038	0.977	0.038	0.987	0.040	0.980
DA3	1.21B		0.173	0.730	0.046	0.974	0.053	0.974	0.044	0.974
Ours (DA3)	1.21B		0.156	0.743	0.043	0.976	0.045	0.978	0.036	0.983

Table 3: Camera pose estimation on Sintel, Bonn and Dycheck. We report ATE, and RPE for translation (Trans) and rotation (Rot). Best results are highlighted.

Method	Sintel			Bonn			Dycheck		
	ATE↓	Trans↓	Rot↓	ATE↓	Trans↓	Rot↓	ATE↓	Trans↓	Rot↓
VGGT	0.148	0.044	0.455	0.015	0.010	0.633	0.022	0.007	0.672
Ours (VGGT)	0.061	0.039	0.300	0.010	0.010	0.630	0.015	0.005	0.614
Pi3	0.060	0.069	0.654	0.013	0.010	0.632	0.011	0.004	0.613
Ours (Pi3)	0.058	0.066	0.668	0.013	0.010	0.630	0.010	0.004	0.615
DA3	0.078	0.046	0.336	0.008	0.008	0.626	0.007	0.003	0.619
Ours (DA3)	0.052	0.030	0.205	0.009	0.009	0.628	0.006	0.003	0.610

4.2 Video Depth Estimation

We evaluate our method on video depth estimation in dynamic scenes, which is particularly challenging due to non-rigid motion, moving objects, and temporal inconsistencies. Following prior work [37, 39, 42], we conduct experiments on a diverse set of real-world and synthetic datasets, including Bonn [26], Sintel [3], KITTI [9], and Dycheck [8], and additionally present qualitative results on in-the-wild videos.

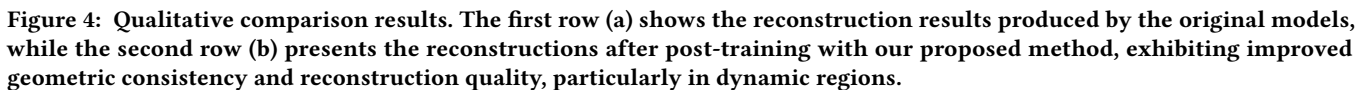
We report Absolute Relative Error (Abs Rel) and threshold accuracy ($\delta < 1.25$) under two alignment settings: (i) scale-only alignment and (ii) joint scale and 3D translation alignment. As shown in Tab. 2, our proposed post-training consistently improves performance across nearly all datasets and backbone architectures, demonstrating the robustness of our approach. Notably, even for Pi3—where the original pixel-shuffle point head is replaced and retrained with fewer parameters—our method yields clear gains. VGGT and DepthAnything3 also benefit consistently from post-training across all evaluated datasets, indicating effective geometric refinement.

As illustrated in Fig. 4, all three models exhibit visibly improved reconstruction quality after post-training, especially in regions with strong object motion. Post-training enhances temporal consistency and reduces dynamic artifacts, leading to more stable and accurate depth predictions. These qualitative results corroborate the quantitative improvements and highlight the effectiveness of our annotation-free post-training strategy for dynamic scene reconstruction.

4.3 Camera Pose Estimation

We evaluate predicted camera poses using two complementary sets of metrics: angular accuracy [37, 42] and distance error [39, 48]. Specifically, we measure performance using the Absolute Trajectory Error (ATE), Relative Pose Error for translation (RPE_{trans}), and Relative Pose Error for rotation (RPE_{rot}) on both synthetic and real-world datasets.

Experiments are conducted on the synthetic outdoor Sintel [3] dataset, as well as the real-world indoor Bonn [26] and DyCheck [8]



and KITTI [9] test sets. These images are processed and used for post-training with our proposed framework.

As shown in Tab. 4, post-training on test-set data consistently improves performance across all evaluated benchmarks. However, the gains are smaller than those obtained when post-training is conducted on large-scale unlabeled video datasets. This indicates

To evaluate the effectiveness of our annotation-free post-training strategy under limited data regimes, we conduct additional experiments on VGGT [37] using images from the Bonn [26], Sintel [3],

Table 4: Effect of annotation-free post-training on VGGT. We report depth and camera pose performance before and after post-training on three benchmarks.

Dataset	Depth (Abs Rel ↓)		Camera Pose (ATE ↓)	
	Before	After	Before	After
Sintel	0.157	0.147	0.048	0.037
Bonn	0.052	0.041	0.015	0.012
KITTI	0.054	0.049	-	-

that, while the proposed method remains effective even with limited data, its benefits scale more substantially with increased data diversity and volume. These results underscore the importance of scaling up annotation-free post-training on rich, large-scale unlabeled video collections to fully unlock the potential of feed-forward 3D reconstruction models.

4.5 Ablation Study

To thoroughly analyze the contribution of each component in our annotation-free post-training framework, we conduct ablation studies on the overall framework effectiveness and its scaling behavior with increasing data.

Component Ablation. We first evaluate the impact of each component on the DyCheck dataset using VGGT as the base model. All variants are tested under the same protocol, reporting both depth estimation accuracy and camera trajectory errors.

Specifically, *w/o all* denotes a variant trained only on datasets with ground-truth annotations (Tab. 1), without leveraging our large-scale unlabeled video data. This results in clear performance degradation, demonstrating the critical importance of annotation-free data for both depth and pose estimation.

We further ablate individual loss terms. Removing the ordinal depth loss ($\mathcal{L}_{order}$) degrades both depth accuracy and camera stability, indicating its importance in preserving relative depth ordering in unconstrained scenes. Similarly, removing the local geometry distillation loss ($\mathcal{L}_{local}$) leads to worse performance, particularly in camera trajectory metrics, highlighting its role in enforcing cross-view geometric consistency.

Overall, the full model achieves the best performance across all metrics, confirming that both large-scale unlabeled data and the proposed geometric supervision losses are essential and complementary.

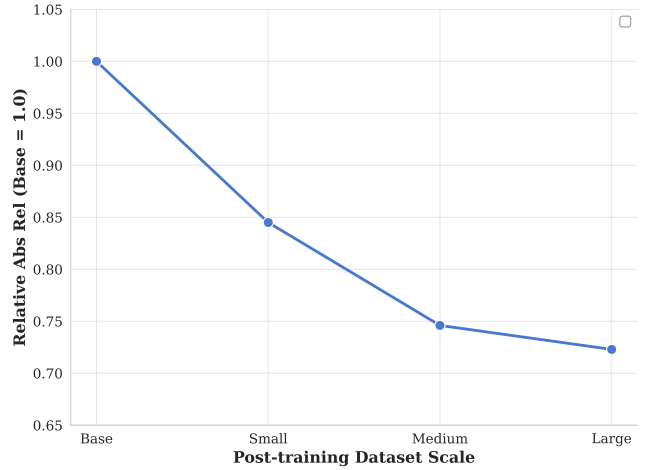
Scaling-Up Analysis. We further study how performance scales with increasing amounts of post-training data. We construct three dataset scales: Small ($\sim 0.2M$ images), Medium ($\sim 1M$ images), and Large ($\sim 5M$ images).

We report the average Absolute Relative Error (Abs Rel) across four datasets (Sintel, Bonn, KITTI, and DyCheck), normalized by the base model (set to 1.0). As shown in Fig. 5, performance improves significantly at smaller scales (0.8452 for Small and 0.7461 for Medium), but saturates at larger scales (0.7229 for Large).

This saturation is expected, as PURE is a post-training framework that does not increase model capacity. Instead, it unlocks the latent

Table 5: Ablation study on Dycheck dataset. We evaluate the impact of different loss terms on depth estimation and camera trajectory accuracy.

Method	Depth		Camera Pose		
	Abs Rel ↓	$\delta < 1.25 \uparrow$	ATE ↓	Trans ↓	Rot ↓
Base Model	0.113	0.852	0.022	0.007	0.672
w/o all	0.084	0.892	0.019	0.006	0.656
w/o $\mathcal{L}_{order}$	0.062	0.957	0.018	0.005	0.635
w/o $\mathcal{L}_{local}$	0.075	0.935	0.016	0.005	0.640
Ours (Full)	0.055	0.976	0.015	0.005	0.614

**Figure 5: Scaling behavior with increasing post-training data. Performance improves significantly as more unlabeled data is introduced, but gradually saturates at larger scales due to the fixed model capacity.**

potential of the base model by exposing it to diverse, in-the-wild data. The results demonstrate that while scaling data is effective, gains are ultimately bounded by the underlying model architecture.

5 Conclusion

We present **PURE**, an annotation-free post-training framework for adapting feed-forward 3D reconstruction models to unconstrained dynamic videos. By distilling relative geometric priors and enforcing correspondence-based pose consistency, PURE significantly enhances depth and camera estimation across state-of-the-art architectures. Ultimately, this work demonstrates that unsupervised post-training is a highly effective, scalable pathway for unlocking robust 3D reconstruction in the wild.

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